

Uniform actual corrector noise at bounded diffusivity

HOCF–RIESZ, Round 013

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Status. This internal synthesis records THM–034, accepted through fresh reconstruction AUD–045 and separate hostile review AUD–046. The complete proofs and version-locked prior particle, full-inverse, domain and energy premises remain in the campaign records. This result controls noise; the integrated cubic drift and the fluctuation theorem remain open.

Statement and fixed conventions

Fix $d \geq 4$, $0 < s < 2$, $0 \leq T, \nu_* < \infty$ and a smooth real terminal test h . Use the unit Haar torus, Fourier characters $e^{2\pi i k \cdot x}$, coefficient-one periodic Riesz potential g , and $K = -\nabla g$. The actual particles start iid Haar independently of their Brownian drivers and satisfy

$$dX_i = \frac{1}{N} \sum_{j \neq i} K(X_i - X_j) dt + \sqrt{2\nu} dW_i.$$

There is no external drift. The reference remains Haar. Let Φ_t be the genuine symmetric terminal-zero full pair inverse with both response slots and the source $J_t(x, y) = K(x - y) \cdot (\nabla f_t(x) - \nabla f_t(y))$, where f is the actual homogeneous Fourier backward test. Set

$$p = s + 2, \quad a = s/p, \quad \theta = 1 - s/d, \quad b = \min(1, \nu^{-1}), \quad \sigma_N^2 = Nb \quad (\nu > 0).$$

The corrector is the literal ordered deleted-pair statistic

$$P_N[\Phi] = \frac{1}{2N^2} \sum_{i \neq j} \Phi(X_i, X_j) - \frac{1}{N} \sum_i \int \Phi(X_i, y) dy + \frac{1}{2} \int \Phi(x, y) dx dy.$$

Its actual scaled expected martingale bracket is

$$Q_N = 2\nu Nb \mathbb{E} \int_0^T \sum_i |\nabla_i P_N[\Phi_t](X_t)|^2 dt.$$

At zero noise define $Q_N = 0$ directly. Uniformly in every $N \geq 2$ and every $0 < \nu \leq \nu_*$,

$$Q_N \leq Cb \min(N^a, \nu^{-s/2})(N^{-\theta} + \nu).$$

Moreover the entire bounded interval satisfies

$$\sup_{0 \leq \nu \leq \nu_*} Q_N \longrightarrow 0,$$

with the sufficient uniform rate

$$Q_N \leq C \left[N^{a-\theta} + N^{-(2-s)/(12p^2)} + N^{-s/(4p(s+4))} + N^{-1/(2p)} + N^{-2/p} \right].$$

Constants depend only on the fixed displayed data and weights. No limiting or boundedness assumption on $\nu N^{2/p}$ is made.

Two occupation estimates for the genuine derivative

Use the fixed periodic multiplicative weights $w_q = w_1^q$, with $w_q(z) = |z|^{-q}$ near zero and $w_1 \geq 1$. For each fixed

$$1 < q < \min(d-2, d/2), \quad q \leq s+1, \quad \eta = s+1-q < 2,$$

the full inverse satisfies

$$\sup_t \sup_{x \neq y} \frac{|\nabla_{x,y} \Phi_t(x, y)|}{w_q(x-y)} \leq C_q \min(N^{\eta/p}, \nu^{-\eta/2}).$$

Write $\mathcal{G} = \nu \Delta_{x,y} + N^{-1} K(x-y) \cdot (\nabla_x - \nabla_y)$. The largest transverse eigenvalue of its drift Jacobian is $2s|x-y|^{-p}/N$; the radial eigenvalue is negative. Let ℓ be this nonnegative upper bound with the fixed smooth remainder included. For a fixed inner radius R , direct differentiation gives

$$(\mathcal{G} + \ell)w_q \leq C_q w_q - 2q(d-2-q)\nu \mathbf{1}_{\{r \leq R\}} w_{q+2} - \frac{2s(q-1)}{N} \mathbf{1}_{\{r \leq R\}} w_{q+p}.$$

Both coefficients are strictly favorable. Annular errors cost a fixed constant because diffusivity is bounded. Stopped nonnegative weighted Itô inequalities give a terminal bound and both occupation bounds, with coefficients ν and $1/N$. Noncollision, Fatou and monotone convergence remove the stops at each fixed N, ν .

The source gradient has weight w_{s+1} . Dividing by w_q and using either occupation moment bounds its time-path integral by $CN^{\eta/p}w_q$ or $C\nu^{-\eta/2}w_q$. Fixed- N higher moments and common local starts from the earlier derivative theorem justify differentiation of expectation *before* these improved uniform estimates are applied. The old constants are not relabeled uniform.

With $D = \operatorname{div} K$, the two homogeneous responses are

$$\mathcal{R}F = - \int F(x+w, y) D(dw) - \int F(x, y+w) D(dw).$$

Their weak derivatives differentiate F , not the singular measure. Separated-singularity convolution gives a bound in the derivative seminorm alone; there is no value-amplitude term. The full differentiated Volterra series has a factorial tail and identifies the same bounded Borel inverse. Equivalently, Gronwall applies to that existing finite- N inverse.

Actual higher-power occupation

For $H_N = N^{-1} \sum_{i < j} g(X_i - X_j)$, the sharp heat-integral floor is $H_N \geq -CN^{s/d}$. Strictly below Coulomb, $D_{c1} \geq cw_p - C$ with $c > 0$, and $p < d$. The exact actual energy identity, initially of mean zero, retains the whole force square and the coefficient $\nu(N-1)$ on pair-divergence occupation. Hence

$$\nu \mathbb{E} \int_0^T w_p(X_1 - X_2) dt \leq C(N^{-\theta} + \nu).$$

No individual pair-force square is extracted from the total-force square. The background compensation contributes the retained $C\nu$ term.

From the exact square to the uniform singular tail

Set $G = \nabla_x \Phi$ and $A(x) = \int G(x, y) dy$. Literal differentiation of the ordered statistic yields

$$v_i = \nabla_i P_N = N^{-2} \left[\sum_{j \neq i} G(X_i, X_j) - NA(X_i) \right].$$

The subtraction is NA , not $(N-1)A$. For any field F with its own row, exchangeability, the actual one-body Haar law and finite-sum Cauchy-Schwarz give

$$Q_N[F] \leq 4\nu b \int_0^T \left[\frac{(N-1)^2}{N^2} \mathbb{E}|F(X_1, X_2)|^2 + \|F\|_{L^2(\text{Haar}^2)}^2 \right] dt.$$

This bounds the complete nonnegative square; no signed triple term is discarded. With $q = p/2$, the derivative estimate gives $|G|^2 \leq C \min(N^a, \nu^{-s/2}) w_p$, proving the direct bound for Q_N . For $0 < \nu \leq N^{-\epsilon}$ it implies

$$Q_N \leq C[N^{a-\theta} + N^{-\epsilon(1-s/2)}].$$

For the remaining interval choose

$$q_- = 1 + s/4, \quad \eta_- = 3s/4, \quad r = p/q_- = \frac{4(s+2)}{s+4} > 2.$$

Radially clip G at magnitude L , writing $G = G^L + R_L$, and recompute the Haar row for each field. Since $w_{q_-}^r = w_p$,

$$|R_L|^2 \leq L^{2-r} |G|^r \leq C \nu^{-\eta_- r/2} L^{2-r} w_p.$$

The actual occupation therefore yields the residual bracket bound

$$\mathcal{T}_N(L) \leq C b \nu^{-\eta_- r/2} L^{2-r} (N^{-\theta} + \nu).$$

The physical ν is already included in the occupation estimate. No additional inverse-noise factor is needed. The separately accepted R10 actual clipping estimate, at $L_N = N^{1/(4p)}$, gives

$$Q_N[G^{L_N}] \leq C(N^{-2/p} + N^{-1/(2p)}).$$

The probability-time-particle Hilbert triangle inequality gives $Q_N \leq 2Q_N[G^{L_N}] + 2\mathcal{T}_N(L_N)$, without orthogonality. Choose $\epsilon = 1/(6p^2)$. On $\nu \geq N^{-\epsilon}$ the tail exponent is

$$\epsilon \eta_- r/2 - (r-2)/(4p) = -s/[4p(s+4)].$$

Combining both intervals proves the stated rate. Since $a - \theta = (s(s+2) - 2d)/(dp) < 0$ for $d \geq 4, 0 < s < 2$, every term decays.

The leading scaled bracket is bounded by $2T \sup \|\nabla f\|_\infty^2$. The expected integral of the absolute scaled cross-variation is bounded by $C\sqrt{Q_N}$ and also vanishes uniformly. No dimension-three, $s = 2$, Coulomb, logarithmic, general-background or unbounded-diffusivity extension is inferred. The old condition $\beta_N N^{2s/d-1} \rightarrow 0$, full subcriticality $\lambda_N = \beta_N N^{s/d-1} \rightarrow 0$, and positive finite criticality remain distinct. The actual cubic drift is the next open assertion.